

## Avant UDS library kick-off

perjantai 8. toukokuuta 2026 12.00

**Kokouksen päivämäärä:** 8.5.2026 12.00

**Sijainti:** Microsoft Teams -kokous

**Linkki Outlook-kohteeseen:** [gagseu4a.rta](#)

**Kutsuviesti**

Hei,

Juteltiin Avant IoT kokouksessa, että optimoidulle UDS kirjastolle olisi käyttöä sekä Avant Power, että Avant Techno puolella. Pidetään tämän osalta suunnittelukokous, ja lähdetään edistämään asiaa.

**Aiheita:**

- UDS kirjaston vaatimukset:
  - Tarvittavat UDS palvelut
  - Arkistettuuri, rapinnat ja toivottu rakenne (vs. udsconan)
  - HW / OS alusta
- Toteutustapa:
  - Kiekkivalintoehdot: Golang, Rust, C
  - Olemassa olevat kirjastot
  - Kehitysympäristö ja versiohallinta
- Aikataulu

Sopisiko tämä aika tehtävän suunnittelulle? (hieman täyttää kalentereissa)

T: Timo

**Timo Tuhkunen**

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**GOFORE**

**Microsoft Teams -kokous**

**Liity:** <https://teams.microsoft.com/join/3893965835140347p=qaTsKbeXZ7CG2sYKjM>

**Kokouksen tunnus:** 389 396 583 514 034

**Tunnuskoodi:** q57H2Hq

[Tarkista linkki](#) | [Järjestelmävillitys](#)

Järjestäjälle: [Kokousohjeet](#)

### Osaillistujat

- [Timo Tuhkunen](#) (Kokouksen järjestäjä)
- [Antti Pasanen](#) (Hyväksymys Outlookissa)
- [Antero Salonen](#) (Hyväksymys Outlookissa)
- [Konsta Siekkä](#) (Hyväksymys Outlookissa)

### Muistiinpanot

Requirements for UDS library

- Required Services (so far):
  - DiagnosticSessionControl
  - ECUReset
  - ReadDataByIdentifier
  - WriteDataByIdentifier
  - SecurityAccess
  - RoutineControl
  - RequestDownload
  - RequestUpload
  - TransferData
  - RequestTransferExit
  - RequestFileTransfer
  - TesterPresent
- No need for DTC support or services.
- Architecture and interfaces could follow udsconan -library.
  - Layered architecture
    - CAN
    - ISO-TP
    - UDS services
  - Backends for UDS layer:
    - Linux: CAN and ISO-TP could be combined for Linux. Primary target.
    - Windows: CAN and ISO-TP need to be separated.
    - DoIP: UDP / IP stack. Maybe Ethernet as backbone instead of CAN. (nice to have)
- HW and OS:
  - Linux and Windows OS must be supported.
  - Library will be compiled for different HW architectures.
  - Little and Big Endian support is needed.
  - Avoid using unaligned memory access.

Implementation:

- Language options: Golang, Rust, C
  - Rust and Golang are favoured.
  - Python bindings supported.
  - Can be compiled for many architectures.
  - C++ libraries can be used.
  - Golang has garbage collection. Rust does not have this.
  - If Rust is selected, then standard library shall be used.
  - Avant Power has some Rust code in Cube product.
  - Using Rust with Bosch ECU is difficult. No compiler available. Antero: No need.
  - Rust is pre-chosen for implementation.
- Available libraries:
  - [GitHub - goibus/automotive\\_diag: Unified Diagnostic Services/UDS \(ISO-14229-1\) KWP2000, ISO-14230, OBD-II \(ISO-9141\) and DoIP \(ISO-15765\) definitions to communicate with the road vehicle ECUs in Rust - GitHub](#)
  - MIT license is OK.
  - Library should be studied as basis for the development.
- Avant GitHub to be used for code and design documents.

Schedule:

- SOP for Avant Power Cube product is in August 2026.
- UDS library is needed for this product.

Development approach.

- Pre-study project:
  - Targets: Requirements, architecture, interfaces, design
  - Workload estimation
  - Hour based work.
- Decision for the implementation project
  - Project costs are needed for the decision.
  - Possible cost sharing and licensing model between Avant and Gofore can be considered.
  - Fixed license fee is desired.
  - Both Avant and Gofore to keep IPRs for the library and its usage.
- Antero: Set-up GitHub repository.
  - Name: UDS library

### Open Item:

- Client or Server implementation. Or should we cover the both?
- Which CAN adapters are used with Windows? This has an impact on the architecture and requirements.
- When is the component needed for CUBE product?
  - SOP was in August.

**Plan for the pre-study:**

- Review existing Rust library. Is any re-use of the code or architecture possible?
  - Define requirements for UDS library, architecture and interfaces.
    - Requirement spec
    - Architecture and interface design spec
    - Documentation in GitHub. Pictures with draw.io.
- Development and testing environment
  - Workload estimation for the library development and testing
    - Possible phasing and prioritization. Linux first.
- Review with Avant
- Agree cost sharing and license model.

**Schedule for pre-study:**

- Review in w24/25 before Konsta's vacation.
- Development could happen in July-August.

**Resourcing:**

- Adilion + support from Konsta & Timo.

**Other:**

- Slack channel for the discussion.
- Hohto project.